TEXTBOOK · FALL 2026

Linear Algebra I

Weeks 1–4

All questions and all formulas from every course file and the assigned textbook sections — vectors, linear combinations and span; dot and cross products; matrices, matrix algebra, determinants; rank and the inverse matrix — each problem with a complete step-by-step solution. Compiled for independent study and Test 1 preparation

Tashkent Technological University · Fall 2026

Sources: course notes, lectures, ISA, TA sessions; G. Strang, Introduction to Linear Algebra (4e) and Linear Algebra and its Applications (4e)

QUESTIONBANK · FORMULACATALOG · FULLSOLUTIONSOCTOBER 2026

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1 Course Map and How to Use This Document

This document is a complete standalone guide for the first four weeks of the Linear Algebra I course in the first semester (Fall 2026). It was manually compiled from each relevant source in the course-files/ folder of the course repository: weekly notes (Notes), lecture presentations, exercise sheets for practical sessions, independent study assignments (ISA), problems from TA sessions, and assigned sections from G. Strang's textbook. Problems are reproduced verbatim; solutions are complete, step-by-step, with verifications.

Week	Linear Algebra I
1	Vectors; linear combinations; span; linear independence
2	Dot product, angles, projections; cross product, areas; matrices: operations
3	Matrix algebra; transposition; matrix multiplication; determinants; scalar triple product
4	Matrix rank; inverse matrix via Gauss-Jordan method

Notation. Matrices are written row-wise: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — first row ab, second row cd; column vector is written as $\begin{bmatrix} x \\ y \end{bmatrix}$, row vector as $\begin{bmatrix} x & y \end{bmatrix}$. Powers and indices are written using ^{} and _: A^{-1}, A^T, a_{11}.

2 Week 1 — Vectors (Linear Algebra I)

2.1 Linear Algebra I: Week 1 Formula Catalog

Vectors are columns in R^n; all of Week 1 is built from addition, scalar multiplication, combinations, span and independence.

$$(u+v)_i = u_i + v_i, \quad (c\,u)_i = c\,u_i \text{ (componentwise operations)} \tag{24}$$

$$c_1 v_1 + \cdots + c_n v_n = \text{linear combination} \tag{25}$$

$$\text{span}\{v_1, \ldots, v_n\} = \text{the set of all such combinations} \tag{26}$$

$$v_1, \ldots, v_n \text{ independent} \Leftrightarrow c_1 v_1 + \cdots + c_n v_n = 0 \Rightarrow c_1 = \cdots = c_n = 0 \tag{27}$$

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}, \quad \hat{x} = \frac{x}{\|x\|} \text{ (unit vector)} \tag{28}$$

$$\text{vector } v \text{ of length } L \text{ along } u: \; v = \frac{L\,u}{\|u\|} \tag{29}$$

$$d(P,Q) = \|Q - P\| \tag{30}$$

2.2 Linear Algebra I, week 1: ISA exercises — questions and complete solutions

$$[][] - 2 - 1$$

Throughout the vectors section, vectors are columns; in part $Au =, v =$. 3 5

$$[][] - 2 - 1$$

1. Let $u = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$**. Compute** $u + v, u - v$**, 2u, −3v, 2u − 3v.**

[Source: ISA LA, week 1, exercise 1]

Solution. Component-wise by (24): $u + v = \begin{bmatrix} -3 \\ 8 \end{bmatrix}$; $u - v = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$; 2u $= \begin{bmatrix} -4 \\ 6 \end{bmatrix}$; −3v $= \begin{bmatrix} 3 \\ -15 \end{bmatrix}$; 2u − 3v $= \begin{bmatrix} -4 & + & 3 \\ 6 & + & 15 \end{bmatrix} = \begin{bmatrix} -1 \\ 21 \end{bmatrix}$.

2. Given $a = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$**; find** $a + b, a - b$**, 3a + 2b.**

$$\begin{bmatrix} 5 & -1 \\ 6 & 12 \end{bmatrix}$$

[Source: ISA LA, week 1, exercise 2]

Solution. $a+b = \begin{bmatrix} -1 & + & 2 \\ 4 & & 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $a-b = \begin{bmatrix} -1 & & 2 \\ 4 & + & 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$; 3a + 2b $= [3\cdot(-1)+2\cdot 2;$ $3 \cdot 4 + 2 \cdot (-2)] = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$.

$$\begin{bmatrix} 2 & 6 & 12 \\ -4 & 8 & \end{bmatrix} \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

3. Find x**, if** $x + \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}$**.**

[Source: ISA LA, week 1, exercise 3]

Solution. Subtract the known vector from both sides: $x = \begin{bmatrix} 7 & & 2 \\ 3 & + & 4 \\ -2 & & 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ -3 \end{bmatrix}$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ -2 \end{bmatrix}$$

4. Find x**, if 3x − 2u** $= v$**, where** $u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$**.**

[Source: ISA LA, week 1, exercise 4]

Solution. Isolate: 3x $= v+$ 2u $= \begin{bmatrix} 8 & & 4 \\ 2 & + & 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$, hence $x = [\frac{4}{3}$; 4/3]. Scalar multiplication is distributive over addition, so dividing a vector by 3 means dividing each component by 3.

5. Compute the Euclidean length of the vectors $u = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$**.**

[Source: ISA LA, week 1, exercise 5]

Solution. By (28): $\|u\| = \sqrt{9} + 16 = 5$; $\|v\| = \sqrt{4} + 9 + 36 = 7$; $\|w\| = 0$ (only the zero vector has length zero).

6. Find the unit vector in the same direction as $v =$**, and verify that 4 its length is 1.**

$$[-\frac{4}{5}; \frac{3}{5}]$$

[Source: ISA LA, week 1, exercise 6]

Solution. $\|v\| = \sqrt{16} + 9 = 5$, therefore $\hat{v} = v/5 = [-\frac{4}{5}$; 3/5]. Verification: $\|\hat{v}\| = \sqrt{\frac{16}{25} + \frac{9}{25}} = 1$.

7. Find the vector of length 10 in the same direction as $v = \begin{bmatrix} 3 \\ -8 \end{bmatrix}$**.**

[Source: ISA LA, week 1, exercise 7]

Solution. Here $\|v\| = \sqrt{9} + 64 = \sqrt{73}$, so we scale by $10/\sqrt{73}$: the answer is $(10/\sqrt{73}) \cdot v = [30/\sqrt{73}; -80/\sqrt{73}]$ — indeed $\|(10/\sqrt{73}) \cdot v\| = 10$ by the homogeneity of the $norm$.

$$\begin{bmatrix} 6 \\ -16 \end{bmatrix}$$

indeed $\|2\text{v}\| = 2 \cdot 5 = 10$ by the homogeneity of the $norm$.

8. Using vector subtraction and length, find the distance between $P = (1, -2, 3)$ **and** $Q = (5, 1, -3)$**.**

[Source: ISA LA, week 1, ex. 8]

Solution. $PQ = Q - P = \begin{bmatrix} 5 & & 1 \\ 1 & + & 2 \\ -3 & & 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$, therefore $d(P, Q) = \|PQ\| = \sqrt{16} + 9 + 36 = \sqrt{61}$.

$$[5 - 1; 1 - (-2); -3 - 3]$$

$$\begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$$

. Therefore $d(P,\,Q) = \|PQ\| = \sqrt{4^2} + 3^2 + (-6)^2 = \sqrt{16} + 9 + 36 = \sqrt{61}$.

9. Let $v_1 =, v_2 =$**. Compute** $2v_1 + v_2,\ -v_1 + 3v_2,\ 4v_1 - 2v_2,\ 0v_1 + 5v_2$**. 2 3**

$$[][][][][]1 - 1 - 1 - 24 + 26$$

[Source: ISA LA, week 1, ex. 9]

Solution. $2v_1 + v_2 =$

$$\begin{bmatrix} 2 & 2 \\ 6 & 3 \end{bmatrix}$$

; $-v_1 + 3v_2 =$

$$\begin{bmatrix} -1 & 3 \\ -3 & 9 \end{bmatrix}$$

; $4v_1 - 2v_2 =$

$$\begin{bmatrix} 4 & -2 \\ 12 & -6 \end{bmatrix}$$

; $0v_1 + 5v_2 =$ (the zero scalar completely eliminatesv_1).

$$\begin{bmatrix} 0 & 5 \\ 0 & 15 \end{bmatrix}$$

$5v_2 =$

$$\begin{bmatrix} 1 & -1 & 7 \\ -1 & -2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 6 \\ 0 & 0 \end{bmatrix}$$

.

10. Find c_1, c_2 **such that** $c_1 + c_2 =$**; check by substitution. 1 2 1**

[Source: ISA LA, week 1, ex. 10]

Solution. Component-wise equations: $c_1 - c_2 = 7$ and $c_1 + 2c_2 = 1$. Subtracting the first from the second:

$$c_1 + c_2 = 1; c_1 - c_2 = 7; 2c_1 + c_2 = 1.$$

$3c_2 = -6$, so $c_2 = -2$ and $c_1 = 5$. Verification:

$$\begin{bmatrix} 5 & & 2 \\ 5 & + & 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

.

11. Determine if $b = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$ **can be written as** $b = c1 \cdot v1 + c2 \cdot v2 + c3 \cdot v3$**, where** $v1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $v3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$**.**

[Source: ISA LA, week 1, ex. 11]

Solution. Components give $c_1 + c_3 = 5$, $c_2 + c_3 = 1$, $c_1 + c_2 = 4$. Adding the first two and subtracting the third: $2c_3 = 2$, so $c_3 = 1, c_1 = 4, c_2 = 0$. The combination exists:$b = 4v_1 + 0 \cdot v_2 + 1 \cdot v_3$. The uniqueness of the solution also shows that the three vectors are independent.

12. Describe geometrically all vectors $c \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix}, c \in R$**; do the vectors** $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ **belong to this** span**?**

[Source: ISA LA, week 1, ex. 12]

Solution. The span is the entire line through the origin in the direction $(-2, 3)$. Checking: $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$ — yes $(c = -2 : -2 \cdot (-2) = 4$ and $-2 \cdot 3 = -6)$; $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$ — yes $(c = 1)$; for $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ we need $-2c = 6$ and $3c = -8, i.e.$ $c = -3$ and $c = -\frac{8}{3}$ — contradiction, so it does not lie on the line.

13. Arev$_1 =, v_2 =$ **linearly independent? Find non-zero** c_1, c_2 **with** $c_1 v_1 + 26 c_2 v_2 = 0$ **and explain geometrically.**

[Source: ISA LA, week 1, ex. 13]

Solution. $v_2 = 3v_1$, so $3v_1 - v_2 = 0$ — a non-trivial dependence $(c_1 = 3, c_2 = -1)$. The vectors are dependent. Geometrically, both lie on the same line through the origin, so together they still only span this line — dependence means that one vector does not give a new direction.

$$\begin{bmatrix} 1 & 0 \\ 1 & \end{bmatrix}$$

14. Are $v1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $v3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ **linearly independent? If dependent, express one through** a **combination of the others.**

[Source: ISA LA, week 1, ex. 14]

Solution. Assume$v_3 = \alpha v_1 + \beta v_2$: components give $\alpha = 1$ (first), $\beta = 1$(second), $\alpha + \beta = 2$(third) — consistent. So$v_3 = v_1 + v_2, a$ non-trivial dependence exists, and the set is dependent. Only two independent directions are present; the span is a plane.

$$\begin{bmatrix}1\\1\\1\end{bmatrix}$$

15. Let $v1 = \begin{bmatrix}1\\2\end{bmatrix}$, $v2 = \begin{bmatrix}3\\1\end{bmatrix}$**. (1) Do they** span$R^2$**? (2) For** $b = \begin{bmatrix}b1\\b2\end{bmatrix}$ **find** $c1, c2$ **with** $c1v1 + c2v2 = b$**. (3) Express** $b = \begin{bmatrix}7\\4\end{bmatrix}$ **using the formula.**

$$21[]b_27$$

c_1, c_2 **with** $c_1v_1 + c_2v_2 = b$**. (3) Express by formula. 4**

[Source: ISA LA, week 1, ex. 15]

Solution. (1)–(2) Solve $c_1 + 3c_2 = b_1$, $2c_1 + c_2 = b_2$. Eliminating: $c_1 = b_1 - 3c_2$, then $2b_1 - 6c_2 + c_2 = b_2$, so $c_2 = (2b_1 - b_2)/5$ and $c_1 = (-b_1 + 3b_2)/5$. The formulas are defined for any (b_1, b_2), so yes, the two vectors spanR^2. $(3)c_1 = (-7 + 12)/5 = 1$, $c_2 = (14 - 4)/5 = 2$: $b = v_1 + 2v_2$.

$$Q = uv = (2 : a = V) + 2Vo.$$

16. Let $u1 = \begin{bmatrix}-1\\2\end{bmatrix}$, $u2 = \begin{bmatrix}-6\\3\end{bmatrix}$ (**as printed in the source; in the standard version** $u2 = 3u1 = \begin{bmatrix}-3\\6\end{bmatrix}$)**. (1) Do they** span$R^2$**? (2) Describe** span**{**$u1, u2$**} geometrically; (3) do the vectors** $\begin{bmatrix}8\\-4\end{bmatrix}$ **and** $\begin{bmatrix}1\\2\end{bmatrix}$ **belong to it?**

$$(1)u_2 = 6u_1, \text{accor}$$

span**{**u_1, u_2 **} geometrically; (3) do they belong to it,? −4 2**

[Source: ISA LA, week 1, ex. 16]

Solution. Comparing the ratios $-6/(-1) = 6$ and 3/2 shows that the printed pair is not proportional, so as printed it spans R^2; in the standard version of the exercise $u2 = 3u1 = \begin{bmatrix}-3\\6\end{bmatrix}$ — that is the one we solve. $(1)u2 = 3u1$, so the span is only span$\{u1\}$. (2) Geometrically it is the single line through the origin in the direction $(-1, 2)$; two collinear vectors never span the plane. (3) For $\begin{bmatrix}8\\-4\end{bmatrix}$: we need $-c = 8$ and 2c $= -4, i.e.$ $c = -8$ and $c = -2$ — a contradiction, so no; for $\begin{bmatrix}1\\2\end{bmatrix}$: we need $-c = 1$ and 2c $= 2, i.e.$ $c = -1$ and $c = 1$ — a contradiction again, so no.

$$y = (\frac{1}{2})\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}x + \begin{bmatrix}1\\0\end{bmatrix}(16)$$

17. Let $v1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $v3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$**. (1) Solve** $c1v1+c2v2+c3v3 = b$ **for an arbitrary** $b = \begin{bmatrix} b1 \\ b2 \\ b3 \end{bmatrix}$**; (2) explain why the three vectors** spanR^3**; (3) express** $b = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$**.**

$$c_1 = (b_1 + b_2 - b_3)/2$$

arbitraryb $= b_2$ **; (2) explain why the three vectors** spanR^3**;**

$$c_1 = (b_1 + b_2 - b_3)/2$$

[Source: ISA LA, week 1, ex. 17]

Solution. (1) Components: $c1 + c3 = b1$, $c2 + c3 = b2$, $c1 + c2 = b3$. Adding all three: $2(c1 + c2 + c3) = b1 + b2 + b3$. Then $c1 = (b1 - b2 + b3)/2$, $c2 = (-b1 + b2 + b3)/2$, $c3 = (b1 + b2 - b3)/2$. $(2)A$ solution exists and is unique for every right-hand side, so the span is all of R^3 (and the vectors are independent). $(3)c1 = (4 - 2 + 6)/2 = 4$, $c2 = (-4 + 2 + 6)/2 = 2$, $c3 = (4 + 2 - 6)/2 = 0$: $b = 4v1 + 2v2$.

$$b = 4v_1 + 2v_2$$

18. Three forces act on a **part:** $F_1 = N, F_2 = N, F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

N**. (1) Find the resultant force; (2) find** F_4 **that brings the part into equilibrium:** $F_1 + F_2 + F_3 + F_4 = 0$**.**

$$\begin{bmatrix} 40 \\ -15 \\ -20 \end{bmatrix}$$

[Source: ISA Linear Algebra, week 1, ex. 18]

Solution. (1) We add component-wise: $F_1 + F_2 + F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

$= 50N$. (2) For equilibrium, we need $F_4 = -(F_1 + F_2 + F_3) =$

$$\begin{bmatrix} -40 \\ 15 \\ 20 \end{bmatrix}$$

N: exactly the opposite vector to the resultant force, so that the total force becomes zero.

2.3 Linear Algebra I, week 1: TA session problems (alternative set) — questions and complete solutions

The TA set contains eight short problems, each with an official solution; we present both, elaborating on the reasoning in detail.

1. **Let** $a = (-1,\ 3,\ 2)$, $b = (4,\ 0,\ -2)$**. Compute** $a + b$ **and 3a** $-b$**; find the midpoint between** a **and** b**.**

   [Source: TA Problems, Week 1, Problem 1]

   Solution. $a + b = (3, 3, 0)$; 3a $-b = (-3 - 4,\ 9 - 0,\ 6 + 2) = (-7,\ 9,\ 8)$. The midpoint is the average $^{a+b} = (\frac{3}{2}, 3_2, 0)$ — the same averaging idea as in the scalar formula for the midpoint of a segment.

2. **Givenp** $+q = (5, 7)$ **andp** $-q = (-1, 3)$**; findp andq.**

   [Source: TA Problems, Week 1, Problem 2]

   Solution. Add the two equations: 2p $= (4, 10)$, sop $= (2, 5)$. Subtract: 2q $= (6, 4)$, soq $= (3, 2)$. Check: $(2, 5)+(3, 2) = (5, 7)$ and $(2, 5)-(3, 2) = (-1, 3)$. The "add and subtract" method is the vector analog of simultaneously solving two scalar equations.

3. **Letu** $= (1,\ 2,\ -1), v = (2,\ -1,\ 0)$**. Compute** $-u+$ **3v; write cu + dv component-wise; can cu + dv equal (5, 0, −1)?**

   [Source: TA Problems, Week 1, Problem 3]

   Solution. $-u+$ 3v $= (-1, -2, 1)+(6, -3, 0) = (5, -5, 1)$. Component-wise cu $+dv = (c+$ 2d, 2c $-d, -c)$. To get (5, 0, −1): the third component forces $c = 1$; then $c+$ 2d $= 5$ gives $d = 2$; and indeed 2c $-d = 0$. Answer: yes, $u+$ 2v $= (5, 0, -1)$.

4. **Letr** $= (2,\ -1), s = (-6,\ 3)$**. Describe all combinations cr + ds; do the vectors** spanR^2**? Find the dependence relation.**

   [Source: TA Problems, Week 1, Problem 4]

   Solution. Since $s = -3$r, any combination equals $(c-$ 3d$)r$ — always a multiple of r. The span is only the line through r: the vectors do not span$a$ plane. The relation 3r $+s = 0$ is nontrivial, which proves dependence (cf. (27)).

5. **Letv** $= (1,\ 0,\ 1), w = (1,\ 2,\ 0)$**. Write cv** $+dw = (x, y, z)$**; find one equation relating** x, y, z**; checkb** $= (4, 6, 1)$**.**

   [Source: TA Problems, Week 1, Problem 5]

   Solution. cv $+dw = (c + d,$ 2d, $c)$. Reading: $z = c$ and $y =$ 2d, so $x = z + \frac{y}{2}$ — the only linear equation describing the plane. For b: right side $= 1 + \frac{6}{2} = 4 = x$, so b lies in the plane; from $z = 1, y = 6$: $c = 1, d = 3$.

6. **Find** c, d **such that** $c(1, 2)+d(3, -1) = (7, 3)$**; write and solve the component-wise equations.**

   [Source: TA Problems, Week 1, Problem 6]

   Solution. Components: $c+$ 3d $= 7$ and 2c $-d = 3$. From the second $d =$ 2c − 3; substituting: $c+$6c$-9 = 7$, so 7c $= 16$, $c = \frac{16}{7}$, $d = 32_7 - 3 = 11_7$. Both components are verified exactly.

7. **Let** $u = (1,\ 1,\ 0), v = (0,\ 1,\ 1), w = (1,\ 2,\ 1)$**. Does** w **add** a **new direction? Describe** span**{**u, v, w**}; give** a **relation equal to zero.**

[Source: TA Problems, Week 1, Problem 7]

Solution. $w = u + v$ (component-wise check: $1+0=1, 1+1=2, 0+1=1$), so w is redundant. The relation $u + v - w = 0$ is non-trivial, so the three vectors are dependent, and the span is just the plane spanned by u and v (they are not scalar multiples).

8. Can two vectors span **all of R^3? Can three vectors** spanR^3 **and at the same time be dependent? Explain through "new directions."**

[Source: TA Problems, Week 1, Problem 8]

Solution. Two vectors span at most a plane — each vector gives at most one new direction, and two directions cannot fill three-dimensional space. Three dependent vectors also span at most a plane: dependence means one of them is a combination of the others and does not add a third direction. To achieve R^3, three independent directions are needed; therefore, three dependent vectors will never spanR^3.

3 Week 2 — Scalar and Vector Products, Introduction to Matrices (Linear Algebra I)

3.1 Linear Algebra I, Week 2: Formula Catalog

Week 2 adds the two products of vector geometry and the first matrix concepts.

$$u \cdot v = u_1 v_1 + \cdots + u_n v_n = u^T v, \quad \cos\theta = \frac{u \cdot v}{\|u\|\,\|v\|} \tag{45}$$

$$u \perp v \Leftrightarrow u \cdot v = 0, \quad \text{proj}_v\, u = \frac{u \cdot v}{v \cdot v}\, v \tag{46}$$

$$u \times v = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}, \quad \|u \times v\| = \|u\|\,\|v\|\,|\sin\theta| \text{ (area)} \tag{47}$$

$$u \times v = -(v \times u), \quad (u \times v) \perp u, v \tag{48}$$

$$A = [a_{ij}] \in \mathbb{R}^{m \times n}, \quad (A+B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\, a_{ij} \tag{49}$$

$$Ax = x_1 a_1 + \cdots + x_n a_n \text{ (column picture)} \tag{50}$$

where a_{j} is column j of A. Sizes must match: Ax exists iff A is m×n and $x \in R^{\text{n}}$, and the output lies in R^{m}. A diagonal matrix $D = diag(d_1, \ldots, d_{\text{n}})$ scales each coordinate: $Dx = (d_1 x_1, \ldots, d_{\text{n}} x_{\text{n}})^{\text{T}}$.

3.2 Linear Algebra I, Week 2: Lecture Classroom Problems — Questions and Full Solutions

[3][]4

1. Let $u = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$, $v = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$**. Compute** $u \cdot v$**; classify the angle.**

[Source: LA Lecture, Week 2, Problem 1]

Solution. $u \cdot v = 3 \cdot 4 + (-2) \cdot 5 = 12 - 10 = 2 > 0$. A positive dot product means $\cos\theta > 0$, so the angle is acute.

2. With $u = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$**: find** $\|u\|$ **and** $dist(u, v)$**.**

[Source: LA Lecture, Week 2, Problem 2]

Solution. $\|u\| = \sqrt{4 + 1 + 4} = 3$. $u - v = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$, so $dist(u,\, v) = \sqrt{9 + 16 + 4} = \sqrt{29}$.

3. Find k **so that** $u = \begin{bmatrix} 2 \\ k \\ -1 \end{bmatrix}$ **is orthogonal to** $v = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$**.**

[Source: LA Lecture, Week 2, Problem 3]

Solution. Orthogonality meansu $\cdot v = 0$: $6 + k - 4 = 0$, so $k = -2$.

4. Find the angle between $u =$ **and** $v =$**. 1 −1**

[Source: LA Lecture, week 2, problem 4]

Solution. $u \cdot v = 1 - 1 = 0$: the vectors are perpendicular, $\theta = 90^\circ$ — no formulas needed, as soon as the dot product becomes zero.

$$\cos\varphi = (u \cdot v)/(||u||||v||)$$

5. Project $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ **onto** $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$**.**

[Source: LA Lecture, week 2, problem 5]

Solution. $u \cdot v = 6, v \cdot v = 2$, therefore $proj_v u = \frac{6}{2} v =$ 3v =. Remainder $31u -$ 3v = satisfies $(u -$ 3v$) \cdot v = 0$: decomposition into parallel and $-1+$ perpendicular components. 1 3

6. Compute $u \times v$ **for** $u = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ **and check orthogonality.**

$$2 \cdot 2 - 0 \cdot (-1)[4]$$

[Source: LA Lecture, week 2, problem 6]

Solution. By (47): $u \times v = [0 \cdot 0 - 2 \cdot 2;\ 2 \cdot (-1) - 2 \cdot 0;\ 2 \cdot 2 - 0 \cdot (-1)] = \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix}$. Check: $(u \times v) \cdot u = -8 + 0 + 8 = 0$ and $(u \times v) \cdot v = 4 - 4 + 0 = 0$ — orthogonal to both inputs.

$$u \times v = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \times \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} = [0 \cdot 0 - 2 \cdot 2;$$

$$2 \cdot (-1) - 2 \cdot 0; 2 \cdot 2 - 0 \cdot (-1)] = \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix} (16)$$

$(-7) = 0$ and $3 \cdot 4 + (-1)(-2) + 2 \cdot (-7) = 0$ — orthogonal to both inputs. 2 1

7. Find the area of the parallelogram with sides $a = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$**.**

$$0 \cdot (-1) - 1 \cdot 3 - 3$$

[Source: LA Lecture, week 2, problem 7]

Solution. $a \times b = [0 \cdot (-1) - 1 \cdot 3;\ 1 \cdot 1 - 2 \cdot (-1);\ 2 \cdot 3 - 0 \cdot 1] = \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$, so the area is $\|a \times b\| =$ $\sqrt{9} + 9 + 36 = \sqrt{54} = 3\sqrt{6}$.

$$2 \cdot 3 - 0 \cdot 1 = 6$$

8. For $A =$**: specify the size, find** a_{13}, a_{22} **and write the columns** $035a_1, a_3$**.**

[Source: LA Lecture, week 2, problem 8]

Solution. A has size 2×3 (rows first). $a_{13} = 4$(row 1, column

$$A = \begin{bmatrix} 2 & 4 & \\ 0 & 3 & 5 \end{bmatrix}$$

$3), a_{22} = 3.a_1 =, a_3 =$. 0 5

$$A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix},$$

$$R = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, H = (94 - R^T B).$$

9. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$**. Find 2A** $-B$**.**

[Source: LA Lecture, week 2, problem 9]

Solution. 2A $= \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; subtract element-wise: 2A $-B = [2 - 4, -4 - 1;\ 6 - (-1),$ $0 - 5] = \begin{bmatrix} -2 & -5 \\ 7 & -5 \end{bmatrix}$.

10. A **has size** $3 \times 2, B$ **is** $2 \times 3, x \in R^2, y \in R^3$**. Which of** Ax, Ay, Bx**, By are defined?**

[Source: LA Lecture, week 2, problem 10]

Solution. Ax: $(3 \times 2)(2 \times 1)OK$, result in R^3. By: $(2 \times 3)(3 \times 1)OK$, result in R^2. Ay is inconsistent $(2 \neq 3)$ and Bx is inconsistent $(3 \neq 2)$: both undefined. Only inner dimensions must match. 4

11. Let $D = diag(2, -1, 3)$, $x = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$**. Find** Dx **and** I**3x.**

[Source: LA Lecture, week 2, problem 11]

Solution. $Dx = [2 \cdot 4; -1 \cdot 5; 3 \cdot (-2)] = \begin{bmatrix} 8 \\ -5 \\ -6 \end{bmatrix}$: each coordinate is scaled by its diagonal entry. I3x $= x = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$: the identity matrix does nothing.

$$3 \cdot (-2) = -6$$

. $I_3 x = x = 5$: the identity matrix does nothing. -2

12. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 4 & 0 & 8 \end{bmatrix}$, $x = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$**. Compute** Ax **by the column picture and through row dot products.**

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 0 & 0 & 8 \end{bmatrix}$$

[Source: LA Lecture, week 2, problem 12]

Solution. Columns: $2 \cdot \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. Rows: $\begin{bmatrix} 1 & 2 \end{bmatrix} \cdot x = 0$, $\begin{bmatrix} -1 & 3 \end{bmatrix} \cdot x = -5$, $\begin{bmatrix} 4 & 0 \end{bmatrix} \cdot x = 8$; stacked together the same $\begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. Both pictures always agree.

13.Linear transformation$L : R^3 \to R^3$is defined by$L(x) = Ax$,

$$\text{where} A = \begin{bmatrix} -1 & 3 & 0 \\ 2 & -1 & 4 \\ 0 & 1 & 2 \end{bmatrix}.$$

13. Given $A =$ **and** $Ax =$**; find** x**. 3 −1 5**

[Source: LA Lecture, week 2, problem 13]

Solution. Components: $x_1 + 2x_2 = 7$, $3x_1 - x_2 = 5$. From the second $x_2 = 3x_1 - 5$;

$$x = [\frac{17}{7}]$$

substitute: $x_1 + 6x_1 - 10 = 7$, so $7x_1 = 17$, $x_1 = \frac{17}{7}$, $x_2 = 16_7$. So $x =$. 16/7

14. Non-zero vectors $u + v$ and $u - v$ are orthogonal. What follows about $\|u\|$ and $\|v\|$?

[Source: LA Lecture, week 2, problem 14]

Solution. Orthogonality gives $0 = (u + v) \cdot (u - v) = \|u\|^2 - \|v\|^2$ (cross terms cancel), so $\|u\| = \|v\|$: equal lengths, and u and v are half-diagonals of a rectangle.

3.3 Linear Algebra I, week 2: ISA exercises — questions and full solutions

1. Let $u = (-3, 2)^T, v = (4, 5)^T$. Compute $u \cdot v$; is the angle acute, right, or obtuse?

[Source: LA ISA, week 2, ex. 1]

Solution. $u \cdot v = -12 + 10 = -2 < 0$, so $\cos\theta < 0$: the angle is obtuse.

2. Let $a = (1, -2, 4)^T$, $b = (3, 0, -1)^T$. Find $a \cdot b$, $\|a\|$, $\|b\|$, $\|a - b\|$.

[Source: ISA LA, week 2, ex. 2]

Solution. $a \cdot b = 3 + 0 - 4 = -1$; $\|a\| = \sqrt{1} + 4 + 16 = \sqrt{21}$; $\|b\| = \sqrt{9} + 0 + 1 = \sqrt{10}$; $a - b = (-2, -2, 5)^T$, $\|a - b\| = \sqrt{4} + 4 + 25 = \sqrt{33}$.

3. Find k such that $u = (2, k, 3)^T$ is orthogonal to $v = (-1, 4, 2)^T$.

[Source: ISA LA, week 2, ex. 3]

Solution. $u \cdot v = -2 + 4k + 6 = 0 \Rightarrow 4k = -4 \Rightarrow k = -1$.

4. Find the exact $\cos\theta$, then θ to the nearest degree, for $u = (3, 1)^{\wedge}\{\}T, v = (-2, 4)^{\wedge}\{\}T$.

[Source: ISA LA, week 2, ex. 4]

Solution. $u \cdot v = 3 \cdot (-2) + 1 \cdot 4 = -6 + 4 = -2$; $\|u\| = \sqrt{10}, \|v\| = \sqrt{20} = 2\sqrt{5}$; $\cos\theta = -2/\sqrt{200} = -2/(10\sqrt{2}) = -\sqrt{2}/10 \approx -0.141$, hence $\theta \approx 98°$.

$$u \cdot v = 3 \cdot (-2) + 1 \cdot 4 = -6 + 4 = -2; ||u|| = \sqrt{10}, ||v|| = \sqrt{20} = 2\sqrt{5};$$
$$\cos\theta = (u \cdot v)/(||u|| \cdot ||v||) =$$

$$(-2)/(\sqrt{10} \cdot 2\sqrt{5}) = (-1)/(\sqrt{50}) = (-1)/(5\sqrt{2}) = -\sqrt{\frac{2}{10}}$$

$-\frac{1}{2} \approx -0.1414$, therefore $\theta \approx 98^\circ$. $\sqrt{5}$

5. Find the projection of $u = (6, 1)^T$ onto $v = (2, -1)^T$; verify that $u - proj$ is orthogonal to v.

[Source: ISA LA, week 2, ex. 5]

Solution. $u \cdot v = 12 - 1 = 11, v \cdot v = 5$: $proj_v u = \frac{11}{5}(2, -1)^T = (22_5, -\frac{11}{5})^T$. Remainder: $u - proj = (\frac{8}{5}, 16_5)^T$; dot product with v: $16_5 - \frac{16}{5} = 0$. Verified.

6. Nonzero p, q satisfy $p \cdot q = \|p\|\|q\|$. What can you conclude about their directions?

[Source: ISA LA, week 2, ex. 6]

Solution. From $(45), \cos\theta = 1$, thus $\theta = 0$: the vectors are directed exactly the same (one is a positive multiple of the other). Equality in the angle formula is a limiting case of the Cauchy–Bunyakovsky–Schwarz idea.

7. For $a = (2,\ 1,\ -1)^T$, $b = (0,\ 3,\ 2)^T$ compute a×b and b×a; verify the sign relationship.

$$[2 \cdot 3 - (-1) \cdot 0; 5 - 5]$$

[Source: ISA LA, week 2, ex. 7]

Solution. $a \times b = (-1) \cdot 0 - 2 \cdot 2 = -4$, and $b \times a = 4 = -(a \times b)$, which

$$[2 \cdot 3 - 1 \cdot 0; 1 \cdot 0 - (-1) \cdot 3; (-1) \cdot 3 - 2 \cdot 2] = \begin{bmatrix} 6 \\ 3 \\ -7 \end{bmatrix}$$

corresponds to (48).

8. Verify that $a \times b$ from exercise 7 is orthogonal to both a and b.

[Source: ISA LA, week 2, ex. 8]

Solution. $a \cdot (a \times b) = 2 \cdot 5 + 1 \cdot (-4) + (-1) \cdot 6 = 10 - 4 - 6 = 0$; $b \cdot (a \times b) = 0 \cdot 5 + 3 \cdot (-4) + 2 \cdot 6 = -12 + 12 = 0$. Both are zero, as guaranteed after (47).

9. Find the area of the parallelogram with sides $u = (1,\ 2,\ 2)^T$, $v = (3,\ -1,\ 0)^T$.

$$[2 \cdot 0 - 2 \cdot (-1); 2 \cdot 3 - 1 \cdot 0; 1 \cdot (-1) - 2 \cdot 3]$$

[Source: ISA LA, week 2, ex. 9]

Solution. $u \times v = 2 \cdot 3 - 1 \cdot 0 = 689$.

$$||u \times v|| = \sqrt{(2 \cdot 0 - 2 \cdot (-1))^2 + (2 \cdot 3 - 1 \cdot 0)^2 + (1 \cdot (-1) - 2 \cdot 3)^2} = \sqrt{4 + 36 + 49}$$
$$\sqrt{89}$$

10. $P = (1,\ 0,\ 1)$, $Q = (3,\ 1,\ 0)$, $R = (0,\ 2,\ 4)$: find the area of triangle PQR.

[Source: ISA LA, week 2, ex. 10]

Solution. Edges from P: $PQ = (2,\ 1,\ -1)^T$, $PR = (-1,\ 2,\ 3)^T$. Cross product: $PQ \times PR = (1 \cdot 3 - (-1) \cdot 2; (-1) \cdot (-1) - 2 \cdot 3; 2 \cdot 2 - 1 \cdot (-1)) = (5,\ -5,\ 5)$, $norm$

$$\sqrt{25 + 25 + 25} = \sqrt{75} = 5\sqrt{3}$$

; triangle area $= (\frac{1}{2}) \cdot 5\sqrt{3} = 5\sqrt{3}/2$ (half the parallelogram area). Swapping u and v changes the sign of the cross product, area doesn't change.

11. Find all real t for which $a = (1, t, 2)^T$ and $b = (3, 6, 6)^T$ satisfy $a \times b = 0$; if none, explain.

[Source: ISA LA, week 2, ex. 11]

Solution. Parallel vectors give zero cross product, so we need $a = cb$: $1 = 3c \Rightarrow c = \frac{1}{3}$; then $t = 6c = 2$; check third: $2 = 6c = 2$ — consistent. So $t = 2$ is the only value (then $a = \frac{1}{3}b$).

12. Let $a = (1, 0, 2)^T$, $b = (2, 1, -1)^T$, $c = (0, 3, 1)^T$. Compute $a \cdot (b \times c)$; interpret the zero value.

$$1 \cdot 1 - (-1) \cdot 34$$

[Source: ISA LA, week 2, exercise 12]

Solution. $b \times c = (-1) \cdot 0 - 2 \cdot 1 = -2$; $a \cdot (b \times c) = 4 + 0 + 12 = 16$.

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix}$$

This is the signed volume of the parallelepiped spanned by three vectors; zero value would mean coplanarity (no 3D volume). Here $|16| = 16$ cubic units. 2 −1 0

13. Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix}$. Specify the size; find a_{12}, a_{23}; write each row and column as a vector.

[Source: ISA LA, week 2, exercise 13]

Solution. Size 2×3 (rows first). $a_{12} = -1$(row 1, column $2), a_{23} = 5$. Rows: $\begin{bmatrix} 2 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 4 & 3 & 5 \end{bmatrix}$; columns: $\begin{bmatrix} 2 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \end{bmatrix}$.

$$[1][]35 - 1$$

14. Let $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$. Compute $A + B, A - B, -$2A, 3A − 2B.

[Source: ISA LA, week 2, exercise 14]

Solution. $A+B = \begin{bmatrix} 6 & 2 \\ -2 & 6 \end{bmatrix}$; $A-B = \begin{bmatrix} -4 & 4 \\ -2 & 2 \end{bmatrix}$; −2A $= \begin{bmatrix} -2 & -6 \\ 4 & -8 \end{bmatrix}$; 3A − 2B $= [3-10, 9+2;$ $-6-0, 12-4] = \begin{bmatrix} -7 & 11 \\ -6 & 8 \end{bmatrix}$.

15. A has size $4 \times 3, B$ — $3 \times 2, x \in R^2, y \in R^3, z \in R^4$. Which of Ax, Ay, Az, Bx, By, Bz are defined and with what output dimensions?

[Source: ISA LA, week 2, exercise 15]

Solution. Ay: $(4\times3)(3\times1)$ — defined, result R^4. Bx: $(3\times2)(2\times1)$ — defined, result R^3. All others $(Ax, Az$, By, $Bz)$ have mismatched inner dimensions and are undefined.

16. Let $D = diag(-2, 3, 1), x = (4, -1, 5)^T$. Compute Dx; describe what D does to each coordinate.

$$Dx = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, x = (2, -1, 3)^T.$$

[Source: ISA LA, week 2, exercise 16]

Solution. $Dx =$ first coordinate scaled by -2 (stretching with reflection), second — by 3, third — by 1 (unchanged). Diagonal matrix — three independent scalings of coordinate axes. $1 - 23, x = (2, -1, 3)^T$.

17. Let $A =$ Compute $Ax(a)$ as a combination of 4 0 −1 columns, (b) through scalar products of rows.

$$1 - 22 + 2 + 9 = 13$$

[Source: ISA LA, week 2, exercise 17]

Solution. $(a)2 - 1 + 3 =$

$$2\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\4\\6\end{bmatrix} - \begin{bmatrix}1\\2\\0\end{bmatrix} + \begin{bmatrix}3\\0\\3\end{bmatrix}$$

$$\begin{bmatrix}4\\2\\9\end{bmatrix}$$

. (b) Row 1: $1 \cdot 2 + (-2)(-1) + 3 \cdot 3 = 13$; row 2: $4 \cdot 2 + 0 \cdot (-1) + (-1) \cdot 3 = 5$; row 3: $2 \cdot 2 + 2 \cdot (-1) + 2 \cdot 3 = 6 - 2 + 6 = 10$.

$$\begin{bmatrix}1 & 4 & 2\\-2 & 0 & 2\\-1 & -1 & 2\end{bmatrix}\begin{bmatrix}2\\-1\\3\end{bmatrix} = \begin{bmatrix}13\\5\\10\end{bmatrix}$$

Same result, as expected.

18. Leta$_1 = (1,\ 2)^T, a_2 = (3,\ -1)^T$. Findx $= (x_1, x_2)^T$ with $x_1a_1 + x_2a_2 = (11,\ 1)^T$.

[Source: ISA LA, week 2, exercise 18]

Solution. Components: $x1 + 3x2 = 11, 2x1 - x2 = 1$. From the second $x2 = 2x1 - 1$; then $x1 + 6x1 - 3 = 11$, so $x1 = 2, x2 = 3$.

19. Let $A = \begin{bmatrix}2 & 4\\-1 & -2\end{bmatrix}$. Describe all x with $Ax = 0$; why are the columns dependent?

[Source: ISA LA, week 2, exercise 19]

Solution. $2x_1 + 4x_2 = 0$ and $-x_1 - 2x_2 = 0$ are the same equation: $2x_1 = -4x_2$. Solutions: $x = t \cdot \begin{bmatrix}-2\\1\end{bmatrix}$, $t \in R$ — the entire solution line. There is a nontrivial combination $-2 \cdot a_1 + 1 \cdot a_2 = 0$, which is the linear dependence of the columns (column 2 equals −2 times column 1).

20. A has size 3×2, with $A\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}2\\-1\\4\end{bmatrix}$ and $A\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}3\\2\\0\end{bmatrix}$. Write down A and

compute $A\begin{bmatrix}-2\\5\end{bmatrix}$ **without row dot products.**

[Source: ISA LA, week 2, exercise 20]

Solution. The two images are the columns of A: $A=\begin{bmatrix}2&3\\-1&2\\4&0\end{bmatrix}$. Linearity: $A\begin{bmatrix}-2\\5\end{bmatrix}=$

$-2\cdot\begin{bmatrix}2\\-1\\4\end{bmatrix}+5\cdot\begin{bmatrix}3\\2\\0\end{bmatrix}=\begin{bmatrix}11\\12\\-8\end{bmatrix}$.

$-2Ae_1+5Ae_2=-2-1+52=12$. 4 0 −8

21. Expand $(u+v)\cdot(u-v)$ **for vectors of the same dimension; prove that** $u+v\perp u-v$ **if and only if** $\|u\|=\|v\|$**.**

[Source: ISA LA, week 2, exercise 21]

Solution. Distributing: $(u+v)\cdot(u-v)=u\cdot u-u\cdot v+v\cdot u-v\cdot v=\|u\|^2-\|v\|^2$ (middle terms cancel). Orthogonality means this equals zero, *i.e.* $\|u\|^2=\|v\|^2$, and since norms are non-negative, $\|u\|=\|v\|$.

22. Self-check: (1) what object gives the scalar product, and what gives the vector product? (2) How does the sign ofu $\cdot v$ **classify the angle? (3) Why does** $a2\times 3$ **matrix multiply** a **vector from** R^3 **but not from** R^2**? (4) How do components ofx act in** Ax**? (5) Which operations from week 2 require equal dimensions, and which require vectors specifically from** R^3**?**

[Source: LA ISA, week 2, self-check]

Solution. (1) Scalar; vector (in R^3). (2) Positive ⇒ acute, zero ⇒ right, negative ⇒ obtuse. (3) The product Ax requires matching inner dimensions: 2×3 times 3×1 works; times 2×1 — no. (4) Like weights in a linear combination of columns $x_1a_1+\cdots+x_na_n$. (5) Dot product, addition and subtraction require equal dimensions; cross product and its geometry (areas, normals) exist only in R^3.

3.4 Linear Algebra I, week 2: TA session (15 problems) — questions and complete solutions

$$[][]-25$$

1.$a=,b=$**: computea** $\cdot b$**; classify the angle. 4 1**

[Source: LA TA, week 2, problem 1]

Solution. $a\cdot b=(-2)\cdot 5+4\cdot 1=-10+4=-6<0$: obtuse.

2. Find the $norm$ **and the unit vector in the direction of** $v=\begin{bmatrix}-2\\1\\2\end{bmatrix}$**.**

[Source: LA TA, week 2, problem 2]

Solution. $\|v\|=\sqrt{4}+1+4=3$; unit vector $\hat{}v=\frac{1}{3}$. 2/3

3. Find the distance between $p=\begin{bmatrix}-2\\4\\3\end{bmatrix}$ **and** $q=\begin{bmatrix}2\\1\\0\end{bmatrix}$**.**

[Source: LA TA, week 2, problem 3]

Solution. $p - q = \begin{bmatrix} -2 & 2 \\ 4 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 3 \end{bmatrix}$, the distance is $\sqrt{16} + 9 + 9 = \sqrt{34}$.

4. Find t, if $\begin{bmatrix} 2 \\ t \\ 4 \end{bmatrix}$ is orthogonal to $\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$.

[Source: LA TA, week 2, problem 4]

Solution. $2 - 3t + 8 = 0 \Rightarrow t = \frac{10}{3}$.

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} t \\ -3 \end{bmatrix} = 0 \Rightarrow 2t + 3 = 0 \Rightarrow$$

$$t = -\frac{3}{2}$$

5. Find $\cos\theta$ for $u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

$$\cos\theta = 1/\sqrt{5}0 = \sqrt{\frac{2}{10}}$$

[Source: LA TA, week 2, problem 5]

Solution. $u \cdot v = 2 \cdot (-1) + 1 \cdot 3 = 1$; $\|u\| = \sqrt{5}, \|v\| = \sqrt{10}$; $\cos\theta = 1/\sqrt{50} = \sqrt{2}/10 \approx 0.141$, $\theta \approx 82°$.

6. Projectu $= (5, -1)^T$ ontov $= (2, 1)^T$; writeu as parallel + orthogonal components.

$$[\frac{7}{5}; \frac{18}{5}]$$

[Source: LA TA, week 2, problem 7]

Solution. $u \cdot v = 9, v \cdot v = 5$: $proj =$

$$(\frac{9}{5}) \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

. Orthogonal component:

$$\begin{bmatrix} 5 \\ -1 \end{bmatrix} - [\frac{14}{5}; \frac{9}{5}]$$

$u - proj =$

$$[\frac{11}{5}; -\frac{18}{5}]$$

. Decomposition:

$$u = [\frac{14}{5}; \frac{9}{5}] + [\frac{11}{5}; -\frac{18}{5}]$$

.

7. Compute $a \times b$ for $a = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$.

$$(-1) \cdot 1 - 3 \cdot 4 - 13$$

[Source: LA TA, week 2, problem 8]

Solution. $a \times b = [(-1) \cdot 1 - 3 \cdot 4;\ 3 \cdot 0 - 2 \cdot 1;\ 2 \cdot 4 - (-1) \cdot 0] = \begin{bmatrix} -13 \\ -2 \\ 8 \end{bmatrix}$.

$$|-48; 0-1|$$

8. $A = (0,\ 0,\ 0)$, $B = (2,\ 1,\ 0)$, $C = (1,\ 0,\ 3)$: find the area of the triangle.

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

[Source: LA TA, week 2, problem 9]

Solution. $AB = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $AC = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$; the cross product $AB \times AC = [1 \cdot 3 - 0 \cdot 0;\ 0 \cdot 1 - 2 \cdot 3;$ $2 \cdot 0 - 1 \cdot 1] = \begin{bmatrix} 3 \\ -6 \\ -1 \end{bmatrix}$, $norm = \sqrt{46}$; area $= \sqrt{46}/2$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{1/2}$$

$= 0 \cdot 1 - 2 \cdot 3 = -6, norm = 46$; area $= \frac{46}{2}$.

$$S = (\frac{1}{2}) \cdot ||(2;1;0) \times (1;0;3)|| = (\frac{1}{2}) \cdot ||(3;-6;$$
$$-1)|| = (\frac{1}{2}) \cdot \sqrt{3^2 + (-6)^2 + (-1)^2} = (\frac{1}{2}) \cdot \sqrt{46}$$

9. For $M =$: size, m_{21}, all columns. 4 3 5

$$\begin{bmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{bmatrix}$$

[Source: LA TA, week 2, problem 10]

Solution. 2×3; $m_{21} = 4$; columns„. 4 3 5

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 3 \\ 5 & 0 & 0 \end{bmatrix}$$

10. Let $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 2 & -2 \end{bmatrix}$**. Find 3A + 2B.**

[Source: LA TA, week 2, problem 11]

Solution. 3A $= \begin{bmatrix} 6 & 3 \\ -3 & 9 \end{bmatrix}$, 2B $= \begin{bmatrix} 0 & 8 \\ 4 & -4 \end{bmatrix}$; sum $= \begin{bmatrix} 6 & 11 \\ 1 & 5 \end{bmatrix}$.

11. Let $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$**. Compute** Ax **as** a **combination of columns.**

[Source: LA TA, week 2, problem 12]

Solution. $1 - 2 + 3 ==$.

$$\begin{bmatrix} 0 & 4 & 1 \\ -2 & 0 & 4 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix} + z \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = [0x + 4y + 1z;$$
$$-2x + 0y + 4z;$$

$$1x + 3y + 2z] = \begin{bmatrix} 4y & + & z & \\ -2x & + & 4z & \\ x & + & 3y & + & 2z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 13 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

12. Leta$_1 =, a_2 =, b =$**. Find allx with** $[a_1 a_2]x = b$**. 2 4 6**

[Source: LA TA, week 2, problem 13]

Solution. Both equations reduce to $x_1 + 2x_2 = 3$ (system is consistent but

$$\begin{bmatrix} 3 & & 2t \\ -7 & 2 & \\ -9 & 4 & \end{bmatrix}$$

redundant, sincea$_2 = 2a_1$ andb $= 3a_1$). General solution:$x =, t \in R$ — t infinitely many weights for the same combination.

13. Without revealing coordinates, explain why $(u \times \mathbf{v}) \cdot (u + v) = 0$**.**

[Source: LA TA, week 2, problem 14]

Solution. The cross product is orthogonal to both its inputs, so it's orthogonal to any of their combinations: $(u \times v) \cdot u = 0$ and $(u \times v) \cdot v = 0$; adding, the dot product withu $+v$ equals $0 + 0 = 0$.

$$[u_1 u_2 u_3] \times [v_1 v_2 v_3] \cdot ([u_1 u_2 u_3] + [v_1 v_2 v_3]) = 0$$

14. A **has size** 2×2**, with** $A \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ **and** $A \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$**. Find** A **and** $A \cdot \begin{bmatrix} 5 \\ -2 \end{bmatrix}$**.**

[Source: TA LA, week 2, problem 15]

Solution. The images of the standard basis vectors are the columns:

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$$

. Then $A \cdot \begin{bmatrix} 5 \\ -2 \end{bmatrix} = 5 \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 4 \\ -13 \end{bmatrix}$; linearity turns the computation into two scalings without scalar products.

4 Week 3 — Matrix algebra, determinants, mixed product (Linear Algebra I)

4.1 Linear Algebra I: formula catalog for week 3

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\, a_{ij}, \quad (A^T)_{ij} = a_{ji}, \quad (A^T)^T = A \tag{65}$$

$$(A + B)^T = A^T + B^T, \quad (cA)^T = cA^T, \quad A^T = A \text{ (symmetric)} \tag{66}$$

$$A \in \mathbb{R}^{m \times n},\ B \in \mathbb{R}^{n \times p} \Rightarrow AB \in \mathbb{R}^{m \times p} \tag{67}$$

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \tag{68}$$

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC \tag{69}$$

$$(AB)^T = B^T A^T, \quad I_m A = A = A I_n \tag{70}$$

$$A\,[\,b_1 \cdots b_p\,] = [\,Ab_1 \cdots Ab_p\,] \text{ (column rule)} \tag{71}$$

$$u^T v = \text{scalar}, \quad uv^T = \text{matrix (outer product)} \tag{72}$$

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \text{ (signed area of the column parallelogram)} \tag{73}$$

$$C_{ij} = (-1)^{i+j} \det M_{ij} \text{ (cofactor)}, \quad \det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \tag{74}$$

where Mij is the minor (delete row i, column j). Expanding along any row or column is allowed; rows or columns with zeros reduce the arithmetic. For triangular matrices the determinant is the product of the diagonal entries. The elementary row-operation rules are collected in Table 1.

Operation on A	**Effect on det A**
Exchange two rows	reverses the sign
Multiply one row by c	multiplies the determinant by c
Add a multiple of one row to another	no change
A zero row, or two proportional rows	determinant is 0
Transpose	$\det(A^T) = \det A$
Product of equally sized square matrices	$\det(AB) = \det A \det B$

Table 1. Determinant rules for square matrices (row versions; column versions hold too).

Scaling the whole $n \times n$ matrix scales all n rows: $\det(\text{cA}) = c^{\text{n}} \det A$. Determinants do not distribute over addition: for $A = B = I_2$, $\det(A + B) = 4$ while $\det A + \det B = 2$.

$$(u \times v) \cdot w = u \cdot (v \times w) = \det[\, u \; v \; w \,] \text{ (scalar triple product)} \tag{75}$$

$$V_{\text{parallelepiped}} = |(u \times v) \cdot w|, \quad h = \frac{|(u \times v) \cdot w|}{\|u \times v\|} \tag{76}$$

$$V_{\text{tetrahedron}} = \frac{1}{6} |\det[\, u \; v \; w \,]|, \quad T = 0 \Leftrightarrow u, v, w \text{ coplanar} \tag{77}$$

4.2 Linear Algebra I: lecture problems for week 3 (15) — questions and complete solutions

$$[][2]1 - 21$$

1. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Compute 2A $-B$ and specify the (2, 1) element; can you add A to $a2 \times 3$ matrix?

[Source: LA Lecture, week 3, problem 1]

Solution. 2A $= \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; 2A $-B = [2-2, -4-1;\ 6-(-1),\ 0-4] = \begin{bmatrix} 0 & -5 \\ 7 & -4 \end{bmatrix}$; the (2, 1) element is $2(3) - (-1) = 7$. Addition with $a2 \times 3$ matrix is undefined — both dimensions must match, not just the total number of elements.

2. Let $C = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \end{bmatrix}$. Write C^T with its size; compute CC^T; is it symmetric?

[Source: LA Lecture, week 3, problem 2]

Solution. $C^T = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 1 & 2 \end{bmatrix}$ (3×2). $CC^T = [1+4+1, 0+6-2;\ 0+6-2, 0+9+4] = \begin{bmatrix} 6 & 4 \\ 4 & 13 \end{bmatrix}$. Diagonal elements are rows, scalar multiplied by themselves; off-diagonal elements are equal — the matrix is symmetric.

3. For C above and $x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ compute Cx both ways.

[Source: LA Lecture, week 3, problem 3]

Solution. Rows: $1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 1 = -1$; $0 + 3 \cdot (-1) + 2 \cdot 1 = -1$; so $Cx = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$.

Columns: $2 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 1 \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. The sizes $(2 \times 3)(3 \times 1)$ give $a2 \times 1$ output.

Columns: $2 - + =$. Dimensions $(2 \times 3)(3 \times 1)$ give output 2×1.

$$C = \begin{bmatrix} 0 & 3 & 2 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

4. With A, B from problem 1 compute AB and BA. Are they equal?

$$\begin{bmatrix} 5 & 1 & 2 \\ -2 & -1 & 1 \\ 1 & -2 & 4 \end{bmatrix}$$

[Source: LA Lecture, week 3, problem 4]

Solution. $AB =$

$$\begin{bmatrix} 6 & 3 \\ 8 & 7 \end{bmatrix}$$

, $BA =$

$$\begin{bmatrix} 8 & 10 \\ 0 & 0 \end{bmatrix}$$

. Different; even when both orders exist, equality is not guaranteed.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

5. Let $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}$**. Predict the sizes of** CD **and** DC**, then compute both.**

[Source: LA Lecture, week 3, problem 5]

Solution. CD: $(2\times 3)(3\times 2) = 2\times 2$; DC: $(3\times 2)(2\times 3) = 3\times 3$. $CD = [2+2-3, 0-2-2;$ $0+3+6, 0-3+4] = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}$; $DC = \begin{bmatrix} 2 & 4 & -2 \\ 1 & -1 & -3 \\ 3 & 12 & 1 \end{bmatrix}$.

6. Let $S =$ **(shift)**, $H =$ (**stretch**), $x =$**. Compute HSx and**

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

SHx; describe each transformation.

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

[Source: LA Lecture, week 3, problem 6]

Solution. $Sx =$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

, then HSx =

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

; also $Hx =$

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

, then SHx =

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

. In HS the shift stretch comes after; in SH the stretch happens first, then the shift is added on top. Same two operations, different order, different result: matrices don't commute.

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

,$v =$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

: computeu$^T v$ anduv^T; check (uv^T)^T =vu^T.

7. Foru $= 20 - 13$

[Source: LA Lecture, week 3, problem 7]

Solution. Inner: $u^T v = 2 + 0 - 3 = -1$ (scalar). Outer: uv^{}{T** =

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 0 & 0 \\ 3 & 0 & -3 \end{bmatrix}$$

(3×3). Transposing element-wise, (uv^T)^T = (vu^T)$_{ij}$, so (uv^T)^T = vu^T. The written matrices are 3×3 transposed to each other.

$$(AB)^T = B^T A^T$$

.

8. Using A, B from problem 1, compute $(AB)^T$ and $A^T B^T$ independently; does $A^T B^T$ match?

$$[4]4 - 76$$

[Source: LA Lecture, week 3, problem 8]

Solution. $AB =$

$$\begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

, so $(AB)^T =$

$$\begin{bmatrix} 3 & -10 \\ 12 & -14 \end{bmatrix}$$

. First

$$\begin{bmatrix} 6 & 3 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

transposed: $A^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix}$$

, $B^T =$

$$\begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$$

; then $A^T B^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

. Computing: $(2 \cdot 1 + (-1)(-2)) = 4$, $(2 \cdot 3 + 0) = 6$

$$\begin{bmatrix} 4 & 6 \\ 28 & -5 \end{bmatrix}$$

$= (AB)^T$, which confirms (70). But $A^T B^T =$

$$\begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

is different: the correct identity changes the order of factors.

9. Let $E = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. Compute $\det E$; find the areas of the parallelogram and triangle spanned by its columns; is orientation preserved?

[Source: LA Lecture, week 3, problem 9]

Solution. $\det E = 3 \cdot 2 - 1 \cdot 1 = 5$. Parallelogram area $= |5| = 5$; triangle area $= \frac{5}{2}$. Orientation is preserved: $\det E > 0$.

10. Compute $\det F$ for $F = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ by cofactors of the first row; check by the first column.

$$\begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

[Source: LA Lecture, week 3, problem 10]

Solution. First row: $\det F = 2\det - 1\det + 0 = 2(6) - 1(-1) = 13$. 0 2 1 2

$$\det F = 3\det\begin{bmatrix}0 & 2\\ 1 & 0\end{bmatrix} - 1\det\begin{bmatrix}1 & 2\\ 0 & 0\end{bmatrix} + 0\det\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = 3(0) - 1(0) + 0(1)$$
$$0.(3.1)$$

First column: $2\det + 1\det = 12 + 1 = 13$ (cofactor sign (3, 1) 0 2 3 1 is $(-1)^{3+1} = +$). Any row or column gives the same value.

11. For $a3 \times 3$ **matrix** $M, \det M = 5$ **is known. Form** N**: swap rows 1,2; multiply row 3 by 2; add** $3 \times$ (**new row 1) to row 2. Find** $\det N, \det(\mathbf{2M}), \det(M^T M)$**.**

[Source: LA Lecture, week 3, problem 11]

Solution. Permutation: sign changes (−5); scaling row 3: another factor of 2; adding a row: no change. $\det N = -2 \cdot 5 = -10$. Scaling all rows: $\det(2\mathrm{M}) = 2^3 \cdot 5 = 40$. Product rule: $\det(M^T M) = \det M^T \det M = 25$.

$$\det(M^T M) = \det M^T \det M = 25.$$

Solution. First the cross product, then the dot product: $u \times v = \begin{bmatrix}-2\\ 1\\ 4\end{bmatrix}$, $(u \times v) \cdot w = 13$.

The parallelepiped volume is 13, the tetrahedron volume is 13/6. Exchanging u and v flips the sign: $(v \times u) \cdot w = -13$ (orientation reverses), the volume magnitudes are the same.

12. Let $u = \begin{bmatrix}2\\ 0\\ 1\end{bmatrix}$, $v = \begin{bmatrix}1\\ 2\\ 0\end{bmatrix}$, $w = \begin{bmatrix}0\\ 1\\ 3\end{bmatrix}$**. Compute** $(u \times v) \cdot w$**, the parallelepiped and tetrahedron volumes for vertices** $0, u, v, w$**; what changes if** u **and** v **are exchanged?**

$$[2 \cdot 2 - 0 \cdot 14]$$

[Source: LA Lecture, week 3, problem 12]

Solution. $u \times v = [1 \cdot 1 - 2 \cdot 0;\ 2 \cdot 0 - 1 \cdot 3;\ 1 \cdot 3 - 2 \cdot 0] = \begin{bmatrix}1\\ -3\\ 3\end{bmatrix}$; dot with w: $1 \cdot 2 + (-3) \cdot 1 + 3 \cdot 4 = 2 - 3 + 12 = 11$. Volume

$$2 \cdot 2 - 0 \cdot 14$$

of parallelepiped 11; of tetrahedron 11/6. Swapping u and v changes the sign to -11, but both volumes remain 11 and 11/6 (absolute values).

$$0 \cdot 0 - 1 \cdot 2 = -2$$

.

13. For $u = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v = \begin{bmatrix}0\\2\\1\end{bmatrix}$, $w = \begin{bmatrix}1\\1\\k\end{bmatrix}$ **find all** k **for which the vectors are coplanar; at that** k **express** w **in terms of** u, v**.**

$$0 \cdot 1 - 1 \cdot 2 - 2$$

Solution. Coplanarity $\Leftrightarrow T = 0$**. Here** $u \times v = [0 \cdot 1 - 1 \cdot 2$**;** $1 \cdot 0 - 1 \cdot 1$**;** $1 \cdot 2 - 0 \cdot 0] = \begin{bmatrix}-2\\-1\\2\end{bmatrix}$**, so** $T = w \cdot (u \times v) = -2 - 1+$ **2k = 2k** $-3 = 0 \Rightarrow k = \frac{3}{2}$**. Comparing the first two coordinates fixes the coefficients:** $w = u + (\frac{1}{2})v = [1$**; 1; 3/2]; the third coordinate agrees exactly at this** k**.**

[Source: LA Lecture, Week 3, Problem 13]

Solution. Coplanarity

$$1 \cdot 2 - 0 \cdot 02$$

means $T = -2 - 1+$ 2k = 2k $-3 = 0 \Rightarrow k = \frac{3}{2}$. Thenw $= \alpha u + \beta v$: the first two coordinates give $\alpha = 1, 2\beta = 1 \Rightarrow \beta = \frac{1}{2}$; checking the third: $1 + 1_2 = 3_2$ — consistent,$w = u + \frac{1}{2}v$.

$$W = U + (\frac{5}{7})V,$$
$$\begin{bmatrix}2 & 0\\0 & 1\end{bmatrix} D = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} (16)$$

14. Let $D = \begin{bmatrix}2 & 0\\0 & 3\end{bmatrix}$, $X = \begin{bmatrix}a & b\\c & d\end{bmatrix}$**. Find all** X **with** $DX = XD$**; what if both diagonal elements of** D **are equal?**

[Source: LA Lecture, Week 3, Problem 14]

Solution. $DX = \begin{bmatrix}2a & 2b\\3c & 3d\end{bmatrix}$, $XD = \begin{bmatrix}2a & 3b\\2c & 3d\end{bmatrix}$. Equality requires $b = c = 0$, with a and d free: precisely diagonal matrices commute with this D. If $D = \lambda I$, then $DX = \lambda X = XD$ for any X.

15. Let $u = \begin{bmatrix}2\\0\\1\end{bmatrix}$, $v = \begin{bmatrix}1\\2\\0\end{bmatrix}$, $w = \begin{bmatrix}0\\1\\3\end{bmatrix}$ **and** $L = \begin{bmatrix}2 & 1 & 0\\0 & 1 & 0\\0 & 0 & -1\end{bmatrix}$**. Without recalculating the edges, find the signed triple product and the volume after applying** L **to every edge; why is the volume factor universal?**

$$\begin{bmatrix}1 & 0 & 3\\0 & 0 & -1\end{bmatrix}$$

find the signed mixed product and volume after applying L **to each edge; why is the volume factor universal?**

[Source: LA Lecture, Week 3, Problem 15]

Solution. Let $U = \begin{bmatrix} u & v & w \end{bmatrix}$; from problem 12, $\det U = 13$. The transformed edges form LU, so $\det(LU) = \det L \det U = (-2)(13) = -26$; new volume 26. The factor $|\det L| = 2$ is independent of U, because $\det(LU) = \det L \det U$ for any U — the determinant product rule has geometric meaning: $|\det L|$ is the volume scaling factor of the transformation.

5 Week 4 — Matrix Rank and Inverse Matrix (Linear Algebra I)

Week 4 of Linear Algebra I covers matrix rank and matrix inversion via Gauss-Jordan elimination, culminating in Test 1 on weeks 1–3. There is no separate exercise sheet for week 4 in the repository, so this section provides $(i)a$ complete catalog of formulas on the two topics, compiled from the assigned Strang material, and (ii)a graded set of practice problems in the style of the course's TA sheets, with full solutions. They use the same mathematical structures as elsewhere in the course, thus serving simultaneously as preparation for Test-1 on the matrix-algebra parts.

5.1 Formula catalog for week 4

6.1.1 Rank and elimination

$$\operatorname{rank} A \le \min(m, n) \tag{106}$$

$$A \in \mathbb{R}^{n\times n}: \ \operatorname{rank} A = n \Leftrightarrow Ax = 0 \text{ has only } x = 0 \tag{107}$$

$$A \text{ invertible} \Leftrightarrow \operatorname{rank} A = n \Leftrightarrow n \text{ pivots} \Leftrightarrow \det A \neq 0 \tag{108}$$

$$A \text{ invertible} \Leftrightarrow Ax = b \text{ has exactly one solution, for every } b \tag{109}$$

6.1.2 The inverse matrix (Strang, Introduction to Linear Algebra 4e, §2.5)

$$AA^{-1} = I = A^{-1}A; \quad x = A^{-1}b \text{ solves } Ax = b \tag{110}$$

$$A^{-1} \text{ exists} \Leftrightarrow \text{elimination gives } n \text{ pivots (row exchanges allowed)} \tag{111}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (ad - bc \neq 0) \tag{112}$$

$$(AB)^{-1} = B^{-1}A^{-1}; \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1} \text{ (reverse order)} \tag{113}$$

$$\operatorname{diag}(d_1, \ldots, d_n)^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \ldots, \frac{1}{d_n}\right) \ (d_i \neq 0) \tag{114}$$

$$Ax = 0 \text{ for some } x \neq 0 \Rightarrow A \text{ is not invertible} \tag{115}$$

$$BA = I,\ AC = I \Rightarrow B = B(AC) = (BA)C = C \text{ (uniqueness)} \tag{116}$$

Interpretation of (115): no matrix can bring 0 back to x.

6.1.3 Gauss-Jordan elimination

$$[\,A \mid I\,] \xrightarrow{\text{elementary row operations}} [\,I \mid A^{-1}\,] \tag{117}$$

The same row operations turn A into I and I into A^{-1}. The row operations are: exchange two rows; multiply a row by a nonzero constant; add a multiple of one row to another. Each corresponds to an elementary matrix E with an easy inverse (undo the operation), which is why $(E_2E_1A)^{-1} = A^{-1}E_1{}^{-1}E_2{}^{-1}$ comes out in reverse order.

5.2 Practice problems for week 4 (rank, inverse, Gauss-Jordan) — questions and full solutions

1. Compute $\text{rank}A$ **for** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 6 & 4 \end{bmatrix}$ **by elimination.**

[Source: LA Practice Week 4, Problem 1]

Solution. Eliminate under the first pivot (1): $R_2 - 2R_1 \to (0, 0, 3)$; $R_3 - 3R_1 \to (0, 0, 1)$. The second column entries vanished, so the second pivot must come from column 3: pivot $= 3$. Eliminate under it: $R_3 - \frac{1}{3}R_2 \to (0, 0, 0)$. Two nonzero pivots: $\text{rank}A = 2$. Note the dependence $\text{col}_2 = 2\text{col}_1$, visible from the start — $a$rank deficiency signal. 2 1

2. Determine if $B =$ **is invertible by three methods: pivots, 4 2 determinant, and nonzero** $null-$**vector.**

[Source: LA Practice Week 4, problem 2]

Solution. (Pivots)$R2 - 2R1 \to (0, 0)$: only one pivot — singular. (Determinant) ad − bc $= 2 \cdot 2 - 1 \cdot 4 = 0$ — singular; both criteria agree.

3. Find the inverse of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$**.**

[Source: LA Practice Week 4, problem 3]

Solution. $\det A = 6 - 5 = 1$, so by $(112)A^{-1} =$. Check: −5 2

$$A^{-1} = (\frac{1}{7}) \cdot \begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix}$$

$$A = [5 \cdot 3 + 3 \cdot (-5)5 \cdot (-1) + 3 \cdot 2; 01] = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

4. Apply Gauss-Jordan to $\begin{bmatrix} A & I \end{bmatrix}$**, to invert** $A =$**. 3 7**

[Source: LA Practice Week 4, problem 4]

Solution.

$$\begin{bmatrix}1 & 2\\ 3 & 7\end{bmatrix} \to R_2 - 3R_1 \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix} \to R_1 - 2R_2 \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix},$$

$$A^{-1} = \begin{bmatrix}-7 & 2\\ 3 & -1\end{bmatrix}$$

So $A - 1 = \begin{bmatrix}7 & -2\\ -3 & 1\end{bmatrix}$; check: $A \cdot A - 1 = \begin{bmatrix}1 & 2\\ 3 & 7\end{bmatrix} \cdot \begin{bmatrix}7 & -2\\ -3 & 1\end{bmatrix} = [7 - 6,\ -2 + 2;\ 21 - 21,$ $-6 + 7] = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = I.$

1

5. Invert by Gauss-Jordan and check that 0 −1 2 the product of pivots equals $\det K$**.**

[Source: LA Practice Week 4, problem 5 (Strang's favorite example)]

Solution. Work on $\begin{bmatrix}K & I\end{bmatrix}$: eliminate with first pivot $2(R_2 + \frac{1}{2}R_1)$, then with second pivot ${}_3(R + 2R)$, arriving at upper triangular form with pivots 2, , . Then clean upward and divide each row by its pivot; the right block becomes ^{4321}

$$K^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix}3 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 3\end{bmatrix}$$

Product of pivots is $2 \cdot \frac{3}{2} \cdot 4_3 = 4 = \det K$, and indeed K^{-1} divides the integer matrix by $\det K$ — the pattern noted after (112). (K is symmetric and tridiagonal, but K^{-1} is dense.)

using A^{-1} from problem 4.

6. Solve $Ax = b$ **for** $A = \begin{bmatrix}1 & 2\\ 3 & 7\end{bmatrix}$, $b = \begin{bmatrix}1\\ 1\end{bmatrix}$**, using the** $A - 1$ **from problem 4.**

[Source: LA Practice Week 4, problem 6]

Solution. $x = A{-}1\mathrm{b} = \begin{bmatrix}7 & -2\\ -3 & 1\end{bmatrix} \cdot \begin{bmatrix}1\\ 1\end{bmatrix} = [7 \cdot 1 - 2 \cdot 1;\ -3 \cdot 1 + 1 \cdot 1] = \begin{bmatrix}5\\ -2\end{bmatrix}.$

$$A^{-1} = (1)/(7) \begin{bmatrix}1 & -2\\ 3 & 1\end{bmatrix}$$

7. Given $A = \begin{bmatrix}1 & 2\\ -3 & 1\end{bmatrix}$ **and** $B = \begin{bmatrix}0 & 2\\ 1 & 1\end{bmatrix}$**; compute** $(AB) - 1$ **without multiplying first, then check.**

[Source: LA Practice Week 4, problem 7]

Solution. By $(113), (AB)-1 = B{-}1\mathrm{A}{-}1$. Here $\det A = 1\cdot 1 - 2\cdot(-3) = 7$, so $A-1 = (\frac{1}{7})\cdot\begin{bmatrix}1 & -2\\ 3 & 1\end{bmatrix}$; $\det B = 0\cdot 1 - 2\cdot 1 = -2$, so $B-1 = (1/-2)\cdot\begin{bmatrix}1 & -2\\ -1 & 0\end{bmatrix} = [-\frac{1}{2}1;\ 1/2\ 0]$. Then $(AB)-1 = B{-}1\mathrm{A}{-}1 = [-\frac{1}{2}1;\ \frac{1}{2}0]\cdot(\frac{1}{7})\cdot\begin{bmatrix}1 & -2\\ 3 & 1\end{bmatrix} = (\frac{1}{14})\cdot\begin{bmatrix}5 & 4\\ 1 & -2\end{bmatrix}$. Check: $AB = \begin{bmatrix}1 & 2\\ -3 & 1\end{bmatrix}\cdot\begin{bmatrix}0 & 2\\ 1 & 1\end{bmatrix} = \begin{bmatrix}2 & 4\\ 1 & -5\end{bmatrix}$, $\det(AB) = -10-4 = -14 = 7\cdot(-2) = \det A\cdot\det B$ — consistent.

$(AB)-1(AB) = [\frac{1}{2}\frac{1}{2};\ \frac{3}{4}\frac{1}{4}]\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = [\frac{1}{2}\frac{1}{2};\ 3/4\ \ 1/4]\ *\ [(-1\ \ 2\ \ 1)/2\ \ (2\ \ 1)/(-2);$ $(-314)/4(21)/(-2)] = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$. Reverse order law confirmed.

8. (Conceptual, Test-1 style.) True or false, with justification: (a) **If** A **is invertible, then** A^T **is invertible.** (b) **If** $Ax = \lambda x$ **with** $x \neq 0$ **and** $\lambda = 0$**, then** A **is not invertible.** (c) **The product of invertible** 3×3 **matrices can be singular.** (d)**rank**$(A) \leq 2$ **for any** 2×3 **matrix.**

[Source: LA Practice Week 4, problem 8]

Solution. (a) True: A has n pivots $\Longleftrightarrow$ A^T has n pivots (row rank = column rank); formally $(A^T)^{-1} = (A^{-1})^T$. (b) True: non-zero x with $Ax = 0\cdot x = 0$ violates (115). (c) False: by (113) the product of invertibles is invertible. (d) True: rank does not exceed the number of rows (each pivot uses one row), so rank $\leq 2 < 3$.

9. (Applied, Test-1 style.) For what value(s) **of** c **is the matrix** $A = \begin{bmatrix}1 & c\\ 4 & 2\end{bmatrix}$ **singular? For the remaining values solve** $Ax = \begin{bmatrix}6\\ 4\end{bmatrix}$**.**

[Source: LA Practice Week 4, problem 9]

Solution. $\det A = c-6$; singular exactly when $c = 6$. Otherwise $A^{-1} =^1$

$$\det A = (c-6)/(4c-20)$$

and$x = A^{-1}b =^1$, for example when $c = 7$:$x = (8,-2)^T$.

$$x = A^{-1}b = (\mathrm{begin}arrayc8-2endarray).$$

10. (Advanced difficulty.) Show that if A **is invertible, then** $\mathrm{rank}(AB) = \mathrm{rank}(B)$ **for any compatible** B**.**

[Source: LA Practice Week 4, Problem 10]

Solution. Multiplication by an invertible matrix does not change the number of pivots: standard reasoning: $\mathrm{rank}(AB) \leq \mathrm{rank}B$ (rows of AB are combinations of rows of B), and applying the same to $B = A^{-1}(AB)$, we get $\mathrm{rank}B \leq \mathrm{rank}(AB)$; both inequalities force equality. Intuition: an invertible factor only stretches/rotates/ shifts spaces, but does not flatten them, so rank cannot be lost or gained.

6 Textbook supplements — additional required reading exercises with full solutions

6.1 Strang, Introduction to Linear Algebra 4e — selected problems solved

1. **(§1.1) Describe all cu for $u = (2, 1)$ and all combinations cu + dv for $u = (2, 1)$, $v = (1, 2)$; when do the combinations cu + dv fill the plane?**
   [Source: Strang Intro 4e, Concept Set §1.1]

   Solution. cu fills the line through (2, 1): every vector on it is a multiple. cu + dv fill the plane if u, v are not collinear.

2. **(§1.2) Compute the dot products: $u \cdot v$ for $u = (1, 2, 1)$, $v = (3, 0, 2)$; find a vector w perpendicular to both.**
   [Source: Strang Intro 4e, Problem Set §1.2]

   Solution. $u \cdot v = 3 + 0 + 2 = 5$**.** A **perpendicular vector is the cross product:** $w = u \times v$** =**

$$\begin{bmatrix}1\\2\\1\end{bmatrix} \times \begin{bmatrix}3\\0\\2\end{bmatrix} = [(2 \cdot 2 - 1 \cdot 0);$$

$$(1 \cdot 3 - 1 \cdot 2); (1 \cdot 0 - 2 \cdot 3)] = \begin{bmatrix}4\\1\\-6\end{bmatrix}$$

   ; check: $w \cdot u = 4 + 2 - 6 = 0$ and $w \cdot v = 12 + 0 - 12 = 0$.

3. **(§1.2) When is $u \cdot v = v \cdot u$ negative and what does that say about the angle? When is it zero?**
   [Source: Strang Intro 4e, Concept §1.2]

   Solution. Negative exactly when $\cos\theta < 0, i.e.$, when the angle is greater than 90° (obtuse); zero when $\theta = 90^\circ$ — perpendicularity. This is the classification used in TA problems of week 2. $21, b = (3, 4)^T$ through A^{-1}.

4. **(§2.5) Solve $Ax = b$ with $A = 12$**

$$[2 - 12; 3 - 42; \frac{2}{3}]$$

   [Source: Strang Intro 4e, Problem Set §2.5]

   Solution. $\det A = 3$: $A^{-1} =^1$; $x = A^{-1}b =^1 =^1 =$.

$$\det A = 3; A = [1 - 12; 3 - 38; 35\frac{5}{3}]; x = A^{-1}b = [\frac{1}{4}; 4;$$
$$2] = [\frac{14}{4}; 4; 2] = [\frac{7}{2}; 4; 2](16)$$

   Check: $2 \cdot \frac{2}{3} + 5_3 = 3$, $2_3 + 2 \cdot 5_3 = 4$.

5. (§2.5) Why does a nonzero solution to $Ax = 0$ rule out invertibility? Give the two-sentence reasoning that the notes call "no matrix can map 0 back to x."

[Source: Strang Intro 4e, Note 4 to §2.5]

Solution. If $Ax = 0$ with $x \neq 0$ and A had an inverse, then $x = A^{-1}Ax = A^{-1}0 = 0$ — contradiction. The zero vector has no "preimage" only when the mapping is one-to-one; the zero vector proves that it is not.

6. (§2.5) If A and C are invertible 3×3 matrices, is $A + C$ invertible? What about AC? What about $A^{-1}C$?

[Source: Strang Intro 4e, Problem Set §2.5]

Solution. $A+C$: not necessarily (for example, $C = -A$ gives 0); sums of invertible matrices can be singular. AC: yes, with $(AC)^{-1} = C^{-1}A^{-1}(113)$. $A^{-1}C$: yes, product of two invertible matrices.

6.2 Strang, Linear Algebra and its Applications 4e, §1.6 (Inverses and Transposes) — key results and worked example

Assigned reading §1.6 adds three items to the catalog: Gauss-Jordan elimination for $A = 211$

$$\det A^{-1} = (1)/(\det A)$$

and the identities -2 7 2 of transposition. 2 1 1

1. Perform Gauss-Jordan elimination on $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ and find $A - 1$.

[Source: Strang LAiA 4e, §1.6, example 1]

Solution. We complete I and eliminate: pivots 2, -8, 1 (after $R_2 - 2R_1, R_3 + R_1$, then $R_3 + \frac{3}{4}R_2$ over the reduced rows). Continuing upward, clearing above pivots and dividing rows by 2, -8, 1, we get

$$A^{-1} = (\frac{1}{8}) \begin{bmatrix} 12 & -5 & -6 \\ 8 & -3 & -2 \\ -1 & 1 & 1 \end{bmatrix}$$

The determinant equals the product of pivots $2(-8)(1) = -16$, and indeed A^{-1} shows division by $\det A$ (accounting for cofactor placement — equivalent to $A_{adj A} =)_{\cdot detA}$

2. State the determinant-transposition identities used in §1.6: $\det A^T$, $\det(AB)$, $\det A^{-1}$.

[Source: Strang LAiA 4e, §1.6]

Solution. $\det A^T = \det A$; $\det(AB) = \det A \det B$ (table 1); and, setting $B = A^{-1}$, $\det(A)\det(A^{-1}) = \det I = 1$, so $1 \det A^{-1} = \det A$, which precisely means "invertible $\Longleftrightarrow$ $\det A \neq 0$", connecting the catalog of determinants from week 3 with the catalog of inverses from week 4.

3. Prove $(A^{-1})^T = (A^T)^{-1}$ and explain why $A^T A$ is symmetric with a nonnegative diagonal.

[Source: Strang LAiA 4e, §1.6]

Solution. Transposing $AA^{-1} = I$, we get $(A^{-1})^T A^T = I$, which means

$$(A^{-1})^T = (A^T)^{-1} \tag{16}$$

precisely: $(A^{-1})^T$ is the inverse of A^T.

$$(A^T)^{-1} = (A^{-1})^T; \text{the matrix} A^T A \text{is symmetric}:$$
$$(A^T A)^T = A^T A$$

its diagonal elements are nonnegative:
$a_{ii} = \Sigma_{k=1}^{n} a_{ki}^2 \geq 0$—sums ofsquares—$a$fact. (17)